Ecology and the environment

25 questions: 8 multiple choice, 17 structured.

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1 · Part 1

Paper 1 & 2 — The organism in the environment — [1]

Which term describes all the populations of different species that live together in the same area?

A population

B community

C habitat

D ecosystem

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2 · Part 1

Paper 2 only — The organism in the environment — [1]

Which statement best describes the term biodiversity?

A the number of individuals of one species living in an area

B the total mass of all the organisms living in an area

C the place where a community of organisms lives

D the variety of different species of organisms living in an area

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3 · Part 1

Paper 1 & 2 — Feeding relationships — [1]

Here is a food chain from the sea.

phytoplankton → zooplankton → herring → seal

Which organism is the secondary consumer?

A herring

B zooplankton

C phytoplankton

D seal

4 · Part 1 Paper 1 & 2 Feeding relationships [1]

Here is a food chain from a rocky shore.

seaweed → limpets → crabs → gulls

The seaweed contains 15 000 kJ of energy.

Approximately how much of this energy is transferred to the crabs?

A 13 500 kJ

B 1 500 kJ

C 150 kJ

D 15 kJ

5 · Part 1 Paper 2 only Cycles within ecosystems [1]

Which type of organism converts ammonium ions into nitrates in the soil?

A nitrogen fixing bacteria

B denitrifying bacteria

C nitrifying bacteria

D decomposers

6 · Part 1 Paper 1 & 2 Human influences on the environment [1]

Which of these gases is a greenhouse gas?

A water vapour

B oxygen

C nitrogen

D hydrogen

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7 · Part 1

Paper 1 & 2 | Human influences on the environment | [1]

Carbon monoxide is an air pollutant.

Why is carbon monoxide harmful to humans?

A. It dissolves in the blood plasma and lowers the pH of the blood.

B. It reacts with glucose in the blood, so cells receive less glucose.

C. It replaces carbon dioxide in the plasma, so carbon dioxide cannot be removed.

D. It combines with haemoglobin, so red blood cells carry less oxygen.

8 · Part 1

Paper 2 only | Human influences on the environment | [1]

Deforestation on a hillside often leads to soil erosion.

Which statement explains why?

A. Burning the trees adds ash to the soil, which makes the soil particles smaller.

B. There are fewer tree roots to hold the soil in place, so rain washes the soil away.

C. More light reaches the ground, so the soil becomes too dry for decomposers.

D. More carbon dioxide in the air makes the soil acidic, so the soil dissolves.

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9 · Part 1

Paper 1 & 2 | The organism in the environment | [5]

In a garden pond there are 35 common frogs, about 300 water snails, several different species of water weed and some dragonfly larvae. The pond also contains water, mud and stones.

(a) Identify one population in the pond. [1]

(b) State what is meant by the term ecosystem. [2]

(c) A tree grows next to the pond. After a few years its branches shade most of the pond.
Explain how this could affect the population size of the water weed. [2]

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10 · Part 1

Paper 1 & 2 The organism in the environment [8]

Leila investigated the population size of daisy plants in two areas of a school field.

- Area X is mown short every week.
- Area Y is never mown, so the grass grows tall.

In each area she placed 10 quadrats. Each quadrat was a square with sides of 0.5 m. She counted the number of daisy plants in each quadrat.

The table shows her results.

Quadrat	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Number of daisy plants in area X	6	9	4	11	7	8	5	10	6	4
Number of daisy plants in area Y	1	0	3	2	0	1	4	2	1	2

(a) Describe how Leila could place the quadrats randomly in each area. [2]

(b) Calculate the mean number of daisy plants per quadrat in area X. [1]

(c) Area X has an area of 600 m^2. The population size can be estimated using the formula:

estimated population size = mean number per quadrat × (total area ÷ area of one quadrat)

Calculate an estimate of the population size of daisy plants in area X. [3]

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(d) Suggest one reason why there were fewer daisy plants in area Y than in area X. [2]

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11 · Part 1

Paper 1 & 2 The organism in the environment [6]

In a greenhouse, pepper plants were attacked by aphids. Ladybirds feed on aphids. The grower recorded the mean number of aphids per plant and the number of ladybirds in the greenhouse every two weeks.

Time / weeks	**0**	**2**	**4**	**6**	**8**	**10**	**12**
Mean number of aphids per plant	20	60	150	210	120	50	40
Number of ladybirds in the greenhouse	4	6	12	30	44	20	10

(a) Describe the changes in the mean number of aphids per plant from week 0 to week 12. Use data from the table in your answer. [3]

(b) Explain the changes in the number of ladybirds between week 4 and week 10. [3]

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12 · Part 1

Paper 2 only

The organism in the environment

[8]

Tomás investigated the distribution of moss and grass from the edge of a wood out into an open field.

He laid a tape measure in a straight line from the edge of the wood (0 m) out into the field. Every 5 m along the tape he placed a quadrat. In each quadrat he estimated the percentage cover of moss and of grass. He also measured the light intensity at each quadrat with a light meter.

Distance from edge of wood / m	**0**	**5**	**10**	**15**	**20**	**25**
Light intensity / lux	1 000	2 500	6 000	12 000	18 000	20 000
Percentage cover of moss / %	65	40	15	5	0	0
Percentage cover of grass / %	5	15	40	70	85	90

(a) Give two variables that Tomás should keep the same each time he uses a quadrat. [2]

(b) Describe the relationship between light intensity and the percentage cover of grass. [2]

(c) Explain why the percentage cover of grass changes along the line. [2]

(d) Tomás concluded that light intensity is the only factor that affects the distribution of moss.

Evaluate this conclusion. [2]

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13 · Part 1

Paper 2 only

The organism in the environment

[7]

Meadows P and Q are next to each other. No fertiliser has ever been added to meadow P. Fertiliser has been added to meadow Q every year for 20 years.

Samira placed 20 quadrats randomly in each meadow. She counted the number of plants of each species in the quadrats.

Plant species	Number of plants in meadow P	Number of plants in meadow Q
perennial ryegrass	40	150
white clover	22	6
buttercup	18	4
daisy	15	0
yarrow	12	0
ribwort plantain	13	0
Total	120	160

(a) Describe how the biodiversity of meadow P differs from the biodiversity of meadow Q. Use data from the table in your answer. [2]

(b) Use the formula:

percentage = (number of perennial ryegrass plants ÷ total number of plants) × 100

Calculate the percentage of the plants counted in meadow Q that were perennial ryegrass. Give your answer to the nearest whole number. [2]

(c) Suggest an explanation for why there are fewer plant species in meadow Q. [2]

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(d) Give a reason why Samira used the same number of quadrats, of the same size, in both meadows. [1]

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14 · Part 1

Paper 1 & 2 | Feeding relationships | [5]

These are the feeding relationships in a food web on a grassy hillside.

- Grass is eaten by rabbits and by grasshoppers.
- Grasshoppers are eaten by frogs and by spiders.
- Frogs are eaten by grass snakes.
- Rabbits are eaten by foxes.

(a) Name the producer in this food web. [1]

(b) Give the only food chain in this food web that contains four organisms. Write it using arrows. [1]

(c) State the trophic level of the grass snake in your food chain from (b). [1]

(d) Explain what could happen to the number of grasshoppers if a disease killed most of the frogs. [2]

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15 · Part 1

Paper 1 & 2 Feeding relationships [6]

In a small orchard, a scientist studied this food chain.

apple trees → caterpillars → blue tits

She estimated the number of organisms and their total dry biomass at each trophic level.

Organism	Number of organisms	Total dry biomass / g
apple trees	10	2 000 000
caterpillars	25 000	7 000
blue tits	20	70

(a) Describe how the shape of a pyramid of numbers for this food chain differs from the shape of a pyramid of biomass for this food chain. [2]

(b) Use the formula:

percentage = (biomass of blue tits ÷ biomass of caterpillars) × 100

Calculate the total dry biomass of the blue tits as a percentage of the total dry biomass of the caterpillars. [2]

(c) Describe how substances in the caterpillars become part of the bodies of the blue tits. [2]

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16 · Part 1 Paper 1 & 2 Feeding relationships [8]

Sheep graze on a field of grass.

- The grass in the field contains 24 000 kJ of energy per m^2 per year.
- The energy that becomes part of new biomass in the sheep is 2 160 kJ per m^2 per year.

(a) Use the formula:

percentage = (energy in new biomass of sheep ÷ energy in grass) × 100

Calculate the percentage of the energy in the grass that becomes part of new biomass in the sheep. [2]

(b) Explain why only a small percentage of the energy in the grass becomes part of new biomass in the sheep. [4]

(c) A farmer can use a field to grow wheat for people to eat, or to grow wheat to feed to sheep that people then eat.

Explain why growing wheat for people to eat provides more energy for people. [2]

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17 · Part 1

Paper 1 & 2 | Cycles within ecosystems | [7]

This question is about the carbon cycle.

(a) Name the process in plants that removes carbon dioxide from the atmosphere. [1]

(b) Describe how the carbon in the cells of a dead leaf is returned to the atmosphere. [3]

(c) In a peat bog the soil is waterlogged and contains very little oxygen. Dead plants build up as a layer of peat over thousands of years.

Suggest why the carbon in the dead plants in a peat bog is not returned to the atmosphere. [2]

(d) Peat can be dug up, dried and burned as a fuel.

State the name of the process that releases the carbon in peat into the atmosphere when it is burned. [1]

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18 · Part 1

Paper 2 only

Cycles within ecosystems

[7]

Clover is a plant that has nitrogen fixing bacteria living in small swellings on its roots. Jamal, a farmer, grows clover in a field.

(a) Describe the role of nitrogen fixing bacteria in the nitrogen cycle. [2]

(b) After harvesting, Jamal ploughs the remains of the clover plants into the soil.

Describe the role of decomposers in releasing nitrogen from the dead clover plants. [2]

(c) Another field on the farm is often waterlogged, so the soil contains very little oxygen.

Explain why crops growing in this waterlogged soil may have fewer nitrates available. [3]

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19 · Part 1

Paper 1 & 2 Human influences on the environment [6]

Lichens are organisms that grow on rocks and tree bark. They absorb substances directly from the air.

Students counted the number of lichen species on trees at different distances from a coal-burning power station. They also used data from monitoring stations to find the mean concentration of sulfur dioxide in the air at each distance.

Distance from power station / km	**1**	**3**	**5**	**8**	**12**
Mean concentration of sulfur dioxide / µg per m^3	95	60	30	12	5
Number of lichen species	1	2	5	9	14

(a) Describe the relationship between the concentration of sulfur dioxide and the number of lichen species. Use data from the table in your answer. [2]

(b) Explain how sulfur dioxide released by the power station could harm organisms living in a lake 20 km away. [3]

(c) Suggest why these data alone do not prove that sulfur dioxide reduces the number of lichen species. [1]

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20 · Part 1

Paper 1 & 2 Human influences on the environment [6]

Many human activities release greenhouse gases into the atmosphere.

(a) Complete the table by giving the greenhouse gas that is mainly released by each activity.

Human activity	Greenhouse gas mainly released
raising cattle for meat and milk	
burning petrol in car engines	
adding nitrogen-containing fertiliser to fields	
leaks from old refrigerators	

[4]

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(b) The concentration of methane in the atmosphere has increased as the world's human population has grown.

Explain why. [2]

21 · Part 1

Paper 1 & 2 Human influences on the environment [7]

(a) Explain how an increase in the concentration of greenhouse gases in the atmosphere can lead to global warming.

Plant species M grows only in cold conditions on mountains. Scientists recorded the lowest altitude at which species M was found on one mountain.

Year	1960	1980	2000	2020
Lowest altitude at which species M was found / m	1 800	1 850	1 950	2 050

[4]

(b) Calculate the increase in the lowest altitude at which species M was found between 1960 and 2020. [1]

(c) Suggest an explanation for the change in the lowest altitude at which species M was found. [2]

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22 · Part 1

Paper 1 & 2 | Human influences on the environment | [7]

Untreated sewage leaked into a river from a pipe. Scientists measured the concentration of dissolved oxygen in the water and counted the mayfly larvae in samples taken at different points along the river.

Position of sample	50 m upstream of pipe	at pipe	100 m downstream	300 m downstream	600 m downstream	1 000 m downstream
Concentration of dissolved oxygen / mg per dm^3	9.5	9.0	3.0	1.5	4.5	8.0
Number of mayfly larvae per sample	35	30	0	0	6	25

(a) Describe how the concentration of dissolved oxygen changes from the pipe to 1 000 m downstream. Use data from the table in your answer. [2]

(b) Explain why the concentration of dissolved oxygen falls between the pipe and 300 m downstream. [3]

(c) Suggest why no mayfly larvae were found at 100 m and 300 m downstream. [1]

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(d) Give one reason why sewage in a river is a risk to the health of people who use the river water. [1]

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23 · Part 1

Paper 1 & 2 Human influences on the environment [8]

Next to a lake, a farmer spreads a fertiliser containing nitrates on a field. A few days later there is very heavy rain. Several weeks later, many dead fish are found in the lake.

(a) Explain how the use of fertiliser on the field could lead to the death of the fish. [6]

(b) Suggest two ways the farmer could reduce the risk of this happening again. [2]

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24 · Part 1

Paper 2 only

Human influences on the environment

[6]

An area of tropical rainforest is cut down and the land is used to grow crops. The area has very heavy rainfall.

(a) After a few years the crops grow poorly because the soil contains fewer mineral ions.

Explain why the soil contains fewer mineral ions after the forest has been cut down. [3]

(b) Explain how cutting down large areas of forest may lead to less rainfall in the area. [3]

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25 · Part 1

Paper 2 only

Human influences on the environment

[8]

In one country, 2 400 km^2 of forest is cleared every year. Each km^2 of this forest removes about 500 tonnes of carbon dioxide from the atmosphere each year.

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(a) Calculate the mass of carbon dioxide that is no longer removed from the atmosphere each year because of the forest cleared in one year.

Give your answer in tonnes, in standard form. [2]

(b) Much of the forest is cleared by burning.

Explain why clearing forest by burning increases the concentration of carbon dioxide in the atmosphere. [4]

(c) State the effect of deforestation on the amount of oxygen released into the atmosphere.
Give a reason for your answer. [2]

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